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Prompt Engineering

Prompt Engineering is the process of creating clear and effective inputs for an AI system to guide it toward the desired output.

Why important

* Improves accuracy by reducing vague or irrelevant responses.
* Saves time by obtaining useful results in fewer attempts.
* Enhances creativity when generating ideas, stories or designs.
* Supports problem solving by guiding the AI through local steps.
* Increases reliability in tasks such as summarization, coding and analysis.

Elements of a good prompt

Task
Role
Audience
Details
Examples

T·R·A·D·E

- Clear instruction - state exactly what you want the AI to do.
- Context - Provide Background information when necessary.
- Specific details - Mention the format, length, audience or style required.
- Constraints - Define limitations such as word count, tone or scope.
- Examples - Give sample input output pairs if a particular pattern is needed.

Prompt engineering techniques

1- Zero shot prompting → Asking the AI to perform a task without examples.

2- Few shot prompting → Providing a few examples before asking the AI to complete a similar task.

3- Chain of thought prompting → Encouraging the AI to explain reasoning step by step.

4- Role prompting → Assigning a role such as "act as a teacher" or "act as a software engineer".

5- Constraint based prompting → Adding rules such as "explain in 100 words".

Ex: Weak prompt

Explain photosynthesis.

Improved prompt

Explain photosynthesis in simple terms for a grade 8 student in about 100 words.

EDUCATION IS THE BEST INVESTMENT

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Applications

- Content generation
- Code generation and debugging
- Question answering and tutoring
- Data analysis and summarization
- Chatbots and virtual assistants.

Best practices

- Use simple & direct language.
- Provide enough context for the AI to understand the task.
- Specify the expected output format.
- Avoid ambiguous or overly broad instructions.

Limitations

- All responses may still contain errors or outdated information.
- Poorly designed prompts can lead to biased, incomplete or irrelevant outputs.
- Prompt engineering doesn't guarantee perfect results.

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